

12 - Refund Tracking and Intelligence Module

Overview

The Refund Tracking and Intelligence module provides end-to-end visibility into the IRS refund pipeline for clients under EFIN 748335, combining WMR polling, TC code analysis, predictive timeline modeling, and anomaly detection.

Refund Pipeline Stages

- [IRS Return Accepted (TC 150)]
- > [WMR: Return Received]

-> [WMR: Refund Approved, TC 846 date set]

-> [WMR: Refund Sent]

- Direct Deposit: 1-5 business days

- Paper Check: 5-21 days

-> [Refund Posted to Client Account]

-> [Status: CLOSED]

WMR Status Mapping

WMR Status	TC Indicator	Typical Timeline	Action
Return Received	TC 150 posted	Day 0	None
Refund Approved	TC 846 scheduled	Day 5-10	Notify client
Refund Sent	TC 846 date reached	Day 10-21	Confirm bank/mail
Need More Information	TC 971 and notice	Variable	Contact IRS
Take Action	TC 570 or 810	Variable	Investigate

Standard Refund Timelines

Filing Method	Direct Deposit	Paper Check
e-File	10-21 days	4-6 weeks

Filing Method	Direct Deposit	Paper Check
Paper File	4-6 weeks	6-8 weeks
Amended (1040-X)	N/A	8-20 weeks
Business Return	3-6 weeks	6-8 weeks

PATH Act note: EIC and ACTC refunds are held until after Feb 15 each filing season.

Refund Intelligence Features

Anomaly Detection

Anomaly	Detection Rule	Alert
Refund delayed > 21 days	TC 150 + 21 days, no TC 846	Critical
Refund frozen	TC 810 present	Critical immediate
Partial refund issued	TC 846 amount < expected	Warning
Refund redirected	TC 846 routing differs from client bank	Critical immediate
Duplicate refund request	Two TC 846 entries in same period	Fraud flag
Address change before refund	TC 014 before TC 846	Fraud flag

Predictive Timeline Model

The model predicts:

- Expected TC 846 posting date
- Expected direct deposit date
- Probability of WMR status change within 48 hours
- Refund hold risk score (0-100)

```
refund_intelligence:
  model: "practitionerhub-refund-v2"
  features:
    - filing_method
    - form_type
    - filing_date
    - agi_band
    - credits_claimed
    - prior_year_hold_flag
    - identity_verified
    - tc_history_pattern
  output:
    - predicted_tc846_date
    - predicted_deposit_date
```

- hold_probability_score
- confidence_interval

Polling Configuration

```
refund_tracking:
  poll_source: "irs_tunnel"
  poll_schedule: "0 */4 * * 1-5"
  path_act_hold_forms:
    - "EITC"
    - "8812"
  path_act_release_date: "Feb 15"
  anomaly_detection: true
  fraud_flag_routing: "audit_protection_engine"
  notification:
    on_status_change: [email, sms, in_app]
    on_refund_issued: [email, sms, in_app]
    on_anomaly: [email, in_app, alert_dashboard]
    on_hold: [email, in_app, preparer_alert]
```

Refund Status Dashboard Metrics

- Total Refunds Pending
- Refunds Approved Today
- Refunds Issued Today
- Average Days to Refund
- Active Anomaly Flags
- PATH Act Holds
- Total Refund Dollar Volume

Refund Risk Score Reference

Score	Risk Level	Meaning	Recommended Action
0-20	Low	Standard processing expected	Monitor normally
21-50	Medium	Minor delay possible	Poll more frequently
51-75	Elevated	Hold/review likely	Proactively contact IRS
76-100	High	Freeze/fraud likely	Immediate preparer action

Integration Points

Module	Integration
IRS Tunnel	WMR status pulls and TC retrieval
TC Sync Engine	Reads TC 570/571/810/811/846
Trigger Engine	Fires on status change/anomaly
Audit Protection	Receives fraud flags

Module	Integration
Client Masterfile	Updates refund status per filing
Notification System	Dispatches client and preparer alerts
TPP Reconciliation	Matches refund outcomes to bank products